MySQL Built-in Functions – Complete Notes

What are Built-in Functions?

Definition:
Built-in functions are **predefined functions provided by MySQL** that perform specific tasks such as calculations, string manipulation, date operations, and data analysis.

Instead of writing complex logic, we can simply call these functions.

Syntax

```
FUNCTION_NAME(argument1, argument2, ...);
```

Example:

```
SELECT UPPER('hello');
```

Output:

```
HELLO
```

Types of Built-in Functions

MySQL built-in functions are mainly divided into:

1. String Functions
2. Numeric Functions
3. Date & Time Functions
4. Aggregate Functions
5. Conversion Functions
6. Control Functions
7. NULL Functions

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1. String Functions

Used to manipulate text.

1. LENGTH()

Returns the number of characters (or bytes, depending on encoding) in a string.

Syntax

```
SELECT LENGTH(string);
```

Example

```
SELECT LENGTH('Computer');
```

Output

```
8
```

UPPER()

Converts text into uppercase.

Syntax

```
SELECT UPPER(string);
```

Example

```
SELECT UPPER('mysql');
```

Output

```
MYSQL
```

LOWER()

Converts text into lowercase.

Example

```
SELECT LOWER('HELLO');
```

Output

```
hello
```

CONCAT()

Joins two or more strings.

Syntax

```
SELECT CONCAT(str1, str2, ...);
```

Example

```
SELECT CONCAT('Akshat',' ','Saini');
```

Output

```
Akshat Saini
```

SUBSTRING()

Extracts part of a string.

Syntax

```
SELECT SUBSTRING(string, start, length);
```

Example

```
SELECT SUBSTRING('DATABASE',2,4);
```

Output

```
ATAB
```

Explanation

```
D A T A B A S E
1 2 3 4 5 6 7 8
```

Starts from position 2 and takes 4 characters.

REPLACE()

Replaces one text with another.

Example

```
SELECT REPLACE('I Love Java','Java','MySQL');
```

Output

```
I Love MySQL
```

TRIM()

Removes spaces from both ends.

```
SELECT TRIM('   Hello   ');
```

Output

```
Hello
```

LEFT()

Returns characters from the left side.

```
SELECT LEFT('Computer',4);
```

Output

```
Comp
```

RIGHT()

Returns characters from the right side.

```
SELECT RIGHT('Computer',3);
```

Output

```
ter
```

REVERSE()

Reverses a string.

```
SELECT REVERSE('MYSQL');
```

Output

```
LQSYM
```

Student Table

```
CREATE TABLE Student(
id INT,
name VARCHAR(50),
marks INT
);

INSERT INTO Student VALUES
(1,'Rahul',80),
(2,'Priya',92),
(3,'Amit',70),
(4,'Neha',95),
(5,'Rohit',60);
```

---

2. Numeric Functions

ABS()

Returns absolute value.

```
SELECT ABS(-15);
```

Output

```
15
```

CEIL()

Rounds upward.

```
SELECT CEIL(5.2);
```

Output

```
6
```

FLOOR()

Rounds downward.

```
SELECT FLOOR(5.9);
```

Output

```
5
```

ROUND()

Rounds to nearest integer.

```
SELECT ROUND(5.6);
```

Output

```
6
```

```
SELECT ROUND(5.3);
```

Output

```
5
```

MOD()

Returns remainder.

```
SELECT MOD(17,5);
```

Output

```
2
```

POWER()

Returns exponent.

```
SELECT POWER(2,5);
```

Output

```
32
```

SQRT()

Square root.

```
SELECT SQRT(64);
```

Output

```
8
```

---

3. Date & Time Functions

NOW()

Current date and time.

```
SELECT NOW();
```

Example Output

```
2026-07-30 12:45:30
```

CURDATE()

Current date.

```
SELECT CURDATE();
```

Output

```
2026-07-30
```

CURTIME()

Current time.

```
SELECT CURTIME();
```

Output

```
12:45:30
```

DAY()

Returns day.

```
SELECT DAY('2026-07-30');
```

Output

```
30
```

MONTH()

Returns month.

```
SELECT MONTH('2026-07-30');
```

Output

```
7
```

YEAR()

Returns year.

```
SELECT YEAR('2026-07-30');
```

Output

```
2026
```

DATEDIFF()

Difference between two dates.

```
SELECT DATEDIFF('2026-08-10','2026-08-01');
```

Output

```
9
```

---

4. Aggregate Functions

Aggregate functions work on **multiple rows** and return **one result**.

Suppose:

ID	Name	Marks
1	Rahul	80
2	Priya	92
3	Amit	70
4	Neha	95
5	Rohit	60

COUNT()

Counts rows.

```
SELECT COUNT(*) FROM Student;
```

Output

```
5
```

SUM()

Adds values.

```
SELECT SUM(marks) FROM Student;
```

Output

```
397
```

AVG()

Average.

```
SELECT AVG(marks) FROM Student;
```

Output

```
79.4
```

MAX()

Highest value.

```
SELECT MAX(marks) FROM Student;
```

Output

```
95
```

MIN()

Lowest value.

```
SELECT MIN(marks) FROM Student;
```

Output

```
60
```

---

5. Conversion Functions

CAST()

Changes one data type into another.

```
SELECT CAST('100' AS SIGNED);
```

Output

```
100
```

CONVERT()

Converts data into another type or character set.

```
SELECT CONVERT('123', SIGNED);
```

Output

```
123
```

---

6. Control Functions

IF()

Works like an if-else statement.

```
SELECT IF(80>=40,'PASS','FAIL');
```

Output

```
PASS
```

CASE

Checks multiple conditions.

```
SELECT
CASE
WHEN 95>=90 THEN 'Excellent'
WHEN 95>=75 THEN 'Good'
ELSE 'Average'
END;
```

Output

```
Excellent
```

---

7. NULL Functions

IFNULL()

Returns another value if the expression is NULL.

```
SELECT IFNULL(NULL,'Not Available');
```

Output

```
Not Available
```

COALESCE()

Returns the first non-NULL value.

```
SELECT COALESCE(NULL,NULL,'Computer',NULL);
```

Output

```
Computer
```

Difference Between Scalar and Aggregate Functions

Scalar Functions	Aggregate Functions
Work on one value	Work on many rows
Return one value for each row	Return one value for the entire group
Examples: UPPER(), LOWER(), ABS(), NOW()	Examples: SUM(), AVG(), MAX(), MIN(), COUNT()

Example

Scalar Function

```
SELECT UPPER(name)
FROM Student;
```

Output

Name
RAHUL
PRIYA
AMIT
NEHA
ROHIT

Aggregate Function

```
SELECT AVG(marks)
FROM Student;
```

Output

AVG(marks)
79.4

Frequently Asked Interview / DSSSB Questions

Q1. What are built-in functions?
Functions already provided by MySQL to perform common operations such as string handling, mathematical calculations, date operations, and data aggregation.

Q2. What is the difference between `COUNT(*)` and `COUNT(column_name)`?

- `COUNT(*)` counts **all rows**, including rows where columns contain `NULL`.
- `COUNT(column_name)` counts only **non-NULL values** in the specified column.

Q3. Which function returns today's date only?

```
SELECT CURDATE();
```

Q4. Which function returns the current date and time?

```
SELECT NOW();
```

Q5. Which function joins multiple strings?

```
CONCAT()
```

Q6. Which function is used to calculate the average?

```
AVG()
```

Q7. Which function finds the highest value?

```
MAX()
```

Q8. Which function replaces part of a string?

```
REPLACE()
```

These are the most commonly used MySQL built-in functions for exams like **DSSSB, NIELIT, university exams, and technical interviews**.